

POLITECNICO DI MILANO







Short term decisions

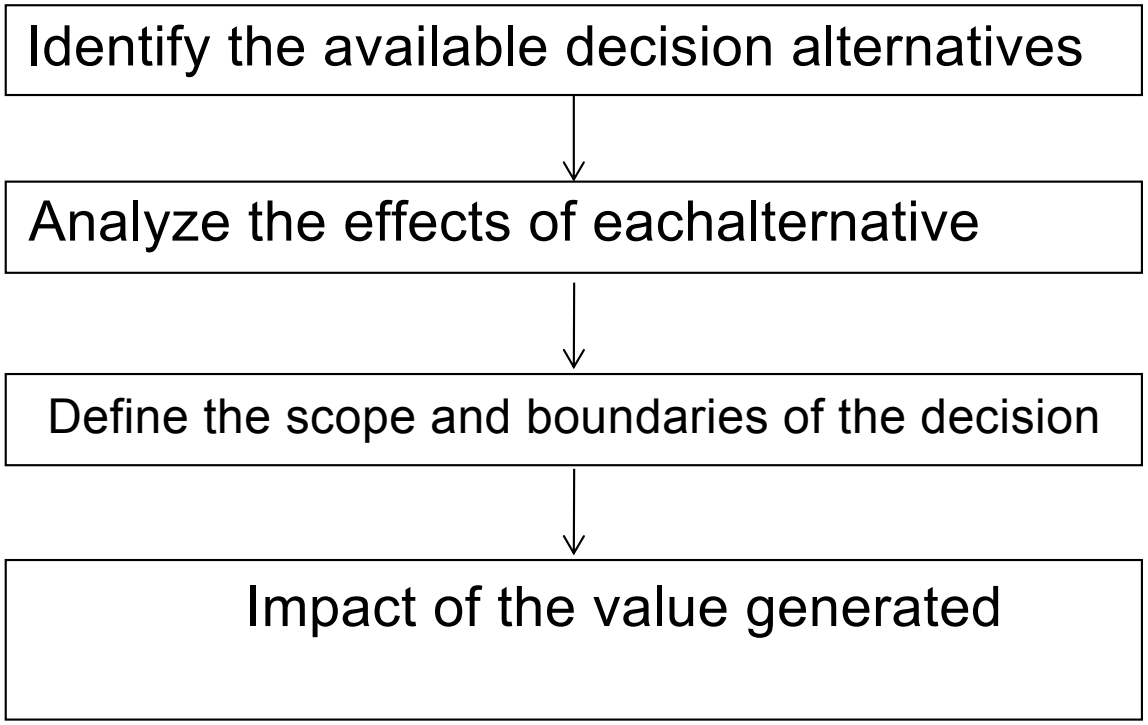
Leonardo Boni, PhD
Dipartimento di Ingegneria Gestionale
Politecnico di Milano
leonardo.boni@polimi.it

Short-Term Decisions

- **Short-Term Decisions**
  Short-term decisions are non-routine decisions that produce effects over a limited time horizon and are taken assuming existing structural resources are fixed.

- **Caracteristics**
  1. They do not change the firm's organizational or production structure
  2. They do not influence the company's strategy
  3. Their economic and financial impact is limited
  4. They do not require investments in physical capital
  5. They are often called tactical decisions, as opposed to strategic decisions

Phase of the decision-making process



Evaluate short term decisions

- In practice, short-term decisions are usually approved by **division managers, rather than by top management**.

  This is because:

- their impact on the company's structure is temporary
- decisions often need to be made quickly

  - For this reason, firms typically adopt simple decision criteria, such as:
    - setting maximum **spending limits in the budget**
    - evaluating the cash flow generated by the decision

Revenues and differential costs

- When evaluating short-term decisions, **cash flows should be calculated by considering only differential revenues and differential costs.**
- Differential revenues (or costs) = the difference between the revenues (or costs) of one alternative and those of another alternative taken as the reference scenario.
  - It is therefore essential to distinguish between:
    - relevant (avoidable) costs
    - irrelevant costs
  - Only relevant costs should be included in the analysis.

Contribution margin

- **Unit contribution margin:** difference between price and variable costs
  $mc = p - c_v$
- **Total contribution margin:** MC = mc***Q** (Q: quantity produced) If differential costs coincide with variable costs, the total contribution margin corresponds to the cash flow generated by the decision.

Types of short term decisions

1. Make or buy decisions

2. Break-even point analysis

Make or buy decisions

1. Make-or-buy decisions involve choosing between:
   a. MAKE → producing a component or product internally
   b. BUY → purchasing it from an external supplier

2. Decision process
   a. Identify the make and buy alternatives
   b. Select one alternative as the base scenario
   c. Calculate differential costs and revenues
   d. Choose the alternative that creates the highest economic value

Make or buy decision: example (1/2)

- A firm must decide whether to produce internally or outsource the production of 100 units of a component.
- In case of internal production (**MAKE**), the firm should buy:
  - raw materials for 50 euros per unit of product component
  - special consulting: 2.000 €

The firm also has:

- two underutilized employees (salary €1,000/month each)
- an underutilized production facility with annual depreciation of €1,000

- Outsourcing case (**BUY**): Supplier price: €80 per unit

Make or buy decision: an example (2/2)

SOLUTION

- Case Base: BUY
- We don't know the revenues from products, we do assume they are identical

- MAKE = MC_{MAKE} = – (50*100)-(2000) + (80*100) =

  – 7.000 + 8000 **> 0**

  We do not consider the salary of the employees (they would have been paid in any case (no differential costs))

  → More convenient decision:
  MAKE

Make or buy decision: some considerations (1/2)

- Opportunity costs should be considered in this decision
- Opportunity cost: the value of the best alternative forgone when choosing a particular course of action.
- Example: If the production facility used to manufacture the component could instead be used to produce other goods, the foregone profit should be included in the analysis.

Make or buy decision: some considerations (2/2)

- Make-or-Buy Decisions in the Long Run
  - Make-or-buy decisions can also arise in a long-term strategic context.
- Example:
  - A firm may decide to acquire one of its suppliers, leading to vertical integration.
  - In such cases, the decision requires:
    - more sophisticated analytical tools
    - capital budgeting techniques
    - **investments evaluation**

Make or buy decision: concluding remarks

The make-or-buy framework presented here has several limitations.

It often ignores **qualitative** factors, such as:

- supplier quality
- reliability of delivery times
- seasonality of component demand
- confidentiality of production knowledge

It may also overlook:

- transaction costs
- potential collaboration strategies

Break-even point

- Break even point help understand how much producing to:
  - cover costs (case 1)
  - obtain a predetermined profitablity (case 2)

- **Break Even Point**
  - Case 1: minimum operating value that considers operating profits equal to costs
    Q_{BE}: EBIT = REVENUES Tot. − COSTS Tot. **= 0**

  - Case 2: minimum operating value to obtaing a predetermined profitability
    Q_{BE}: EBIT = REVENUES Tot. − COSTS Tot. = **Expected profitability > 0**

Break-even point: assumptions

1. Assumptions on **costs**
   - Constant yields
     No economies of scales
     - For example: Learning by doing, efficiency, etc

2. Assumptions on **revenues**
   - Revenues=cash flows. No credit issues
   - No storage management

3. Assumption on **prices**
   - Constant

Break-even point (1/2)

CASE 1:

- Break Even point:

$$Q_{BE} = \frac{C_F}{mc}$$

- Break even point for multi-product firms: weighted average indication of the j-product break even quantity, with respect to the production mix

$$Q_{BEj} = \frac{C_F}{mc_{medio}} * \alpha_j \qquad \alpha_j = \frac{Q_j}{Q_{tot}}$$

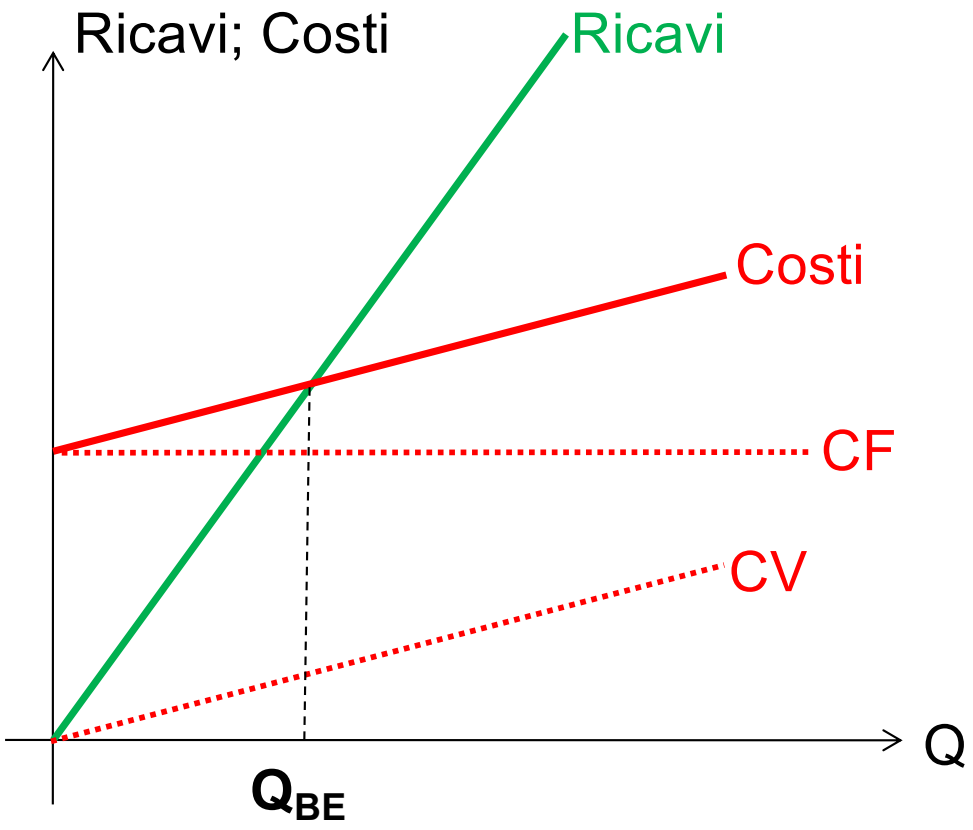
$$mc_{average} = \sum_{j=1}^{N} mc_j * \alpha_j$$

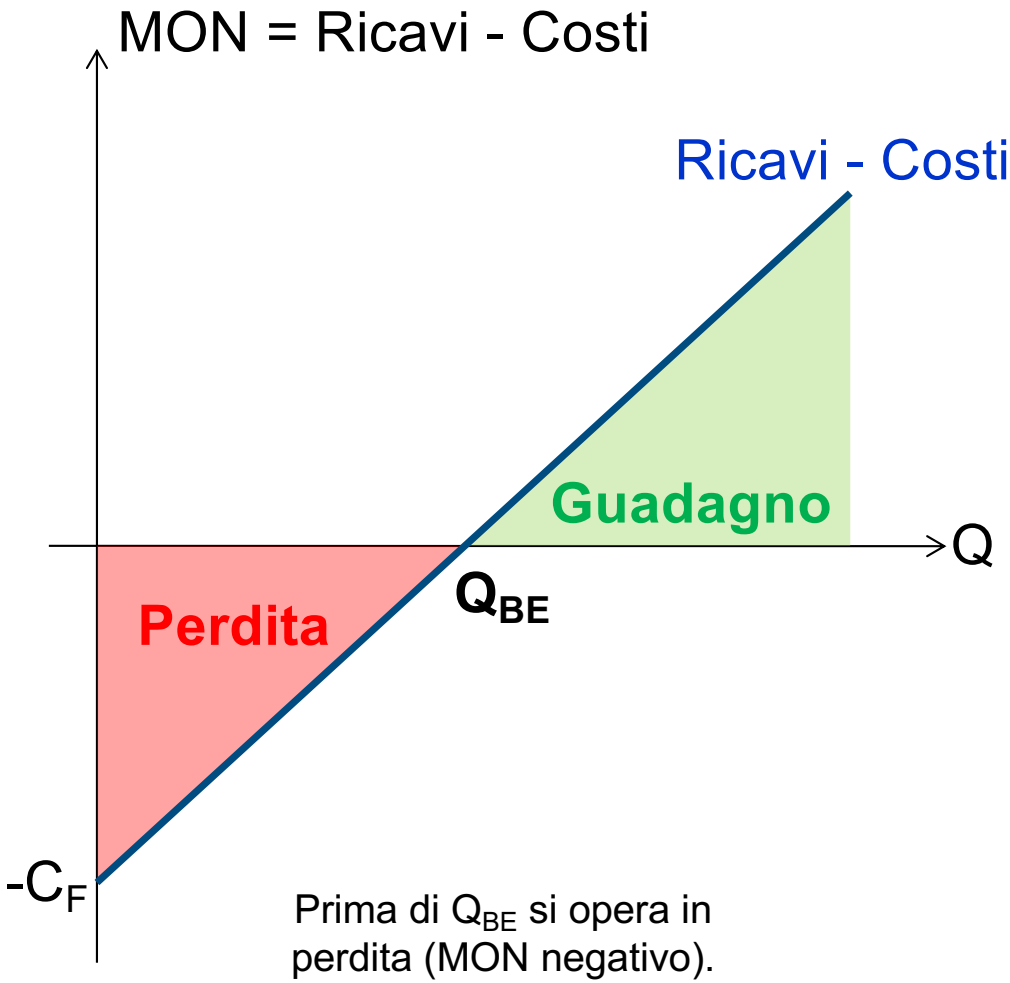
CASE 2:

- Break Even point:

$$Q_{BE} = \frac{\text{Expected profitability} + C_F}{mc}$$

Quantità di break-even (2/2)





Prima di Q_{BE} si opera in perdita (MON negativo). Dopo QBE si opera in profitto (MON positivo)

Interpretation of the break-even point

Break-even point is a quantity

- The break-even quantity indicates:
  the minimum sales volume required to enter the profit zone
  the firm's margin of safety

- Margin of safety:
  $(Q_{max} - Q_{BE})/Q_{max}$
  with Q_{max}: maximum capacity of firm's production
  - A break-even point close to capacity implies high risk, since small drops in demand can quickly lead to losses.

Break-even point analysis: an example (1/2)

- A firm is planning to launch a new product and wants to determine:
  - The sales volume required to cover production costs
  - The sales volume required to achieve a profit of €30,000

- Data
  - Fixed cost (patent): €20,000
  - Variable costs:
    - Materials: €50 per unit
    - Labor: €100 per unit
  - Selling price: €200 per unit

Break-even point analysis: an example (2/2)

Solution

1. Quantities to reach breakeven point

   C_F = 20.000 €

   mc = 200-50-100 = 50 €/unit

   $Q_{BE} = C_F/mc$ = 20.000/50 = **400 units**

2. Quantities to achieve profits of 30.000 €

   C_F = 20.000 €

   Profits expectations = 30.000 €

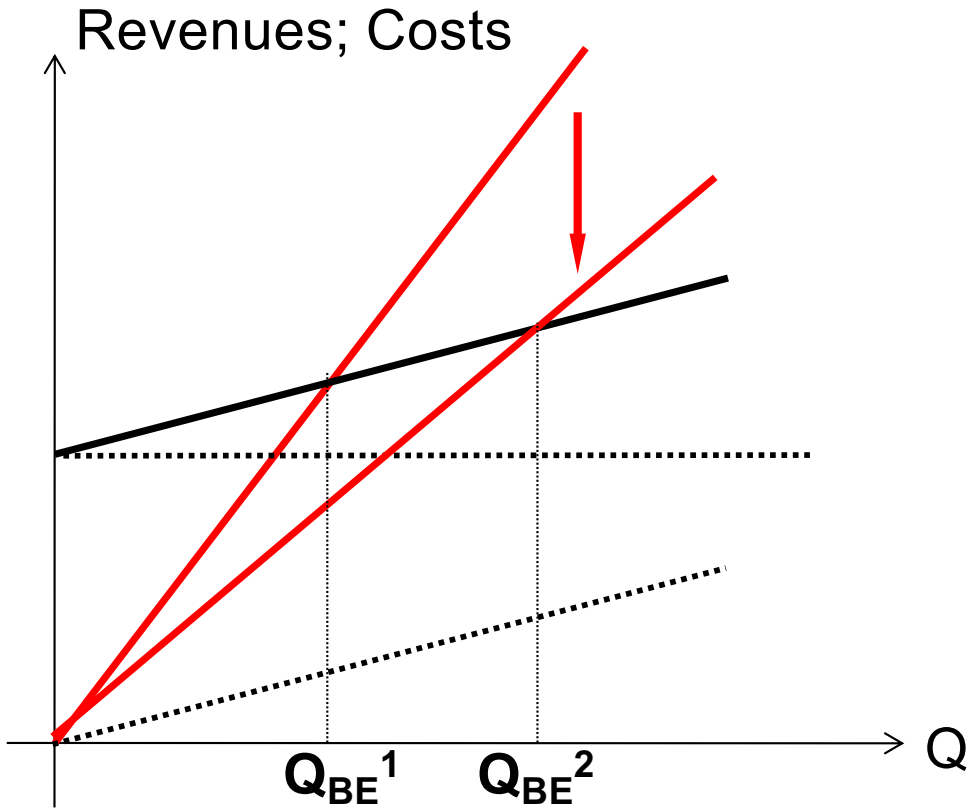
   mc = 200-50-100 = 50 €/unit

   $Q_{BE} = C_F/mc$ = (30.000+20.000)/50 = 50.000/50 = **1.000 units**

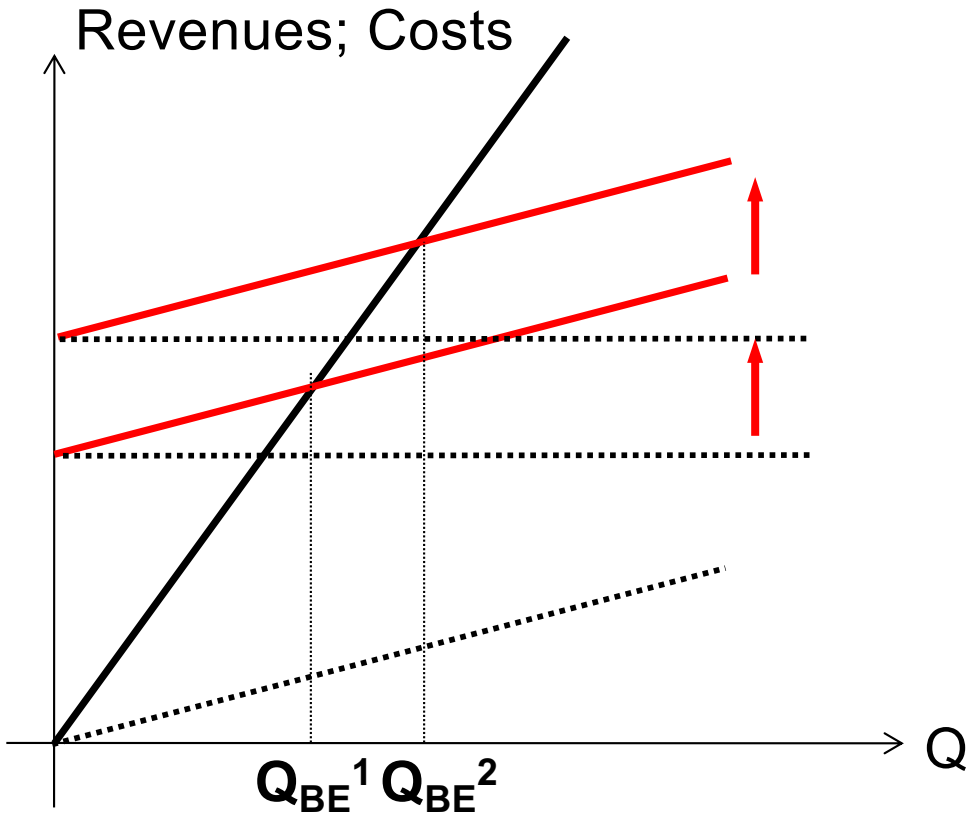
Analysis of the break even point

- Effects of the variation of prices per unit and fixed costs on the break-even point

Reduction of price per unit



Increase of fixed costs



Analysis of the break even point - concluding remarks

In real-world applications, additional cost structures should be considered:

- Semi-fixed costs (e.g., labor contracts, service agreements)
- Upfront fixed costs (e.g., R&D investments before production starts)

Cost structure and risk

Firms with high fixed costs face higher operating risk
This can be measured using the operating leverage index:

Rigidity index: Operating Leverage = Fixed Costs / Total Costs

Implications of high operating leverage

Greater sensitivity to demand fluctuations
Strong incentive to maintain high capacity utilization

Analysis of the break even point - concluding remarks

Strategies to manage high fixed costs

1. Cost flexibilization

Outsourcing activities
Decentralizing production phases
Purchasing external components

2. Process innovation

Improving production efficiency
Increasing automation → reducing labor costs

3. Strategic alliances and acquisitions

Sharing production capacity across firms
Reducing demand volatility risk